

Practical 6

Modeling UML Class Diagrams

Structural and Behavioral aspects | Class Diagram | Elements in Class Diagram | Class | Relationships | Draw Class Diagram

Aim of the Experiment

To model the UML class diagram for a Pharmacy Management System, showing the structural and behavioral aspects of the system.

Introduction

The Pharmacy Management System is designed to automate various tasks involved in managing a pharmacy such as maintaining medicine inventory, managing customer details, generating bills, and handling prescriptions. UML Class Diagrams provide a blueprint for building the structure of the system and capturing the relationship among classes.

Objectives

- To understand and represent the different classes involved in the Pharmacy Management System.
 - To visualize the relationships and interactions between these classes.
 - To graphically represent the system using UML Class Diagram notations.
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After completing this experiment, you will be able to:

- Graphically represent a class, and associations among different classes.
 - Identify the logical sequence of activities undergoing in a system, and represent them pictorially.
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Theory

UML (Unified Modeling Language) is a standardized modeling language used to visualize the design of a system. It helps developers and stakeholders understand, design, and communicate about system structure and behavior.

In object-oriented analysis and design, class diagrams play a crucial role in representing the static structure of the system. They show the system's classes, their attributes, methods, and the relationships among the objects.

For a Pharmacy Management System, class diagrams help in modeling real-world entities like medicines, customers, prescriptions, and billing processes, enabling better planning and efficient implementation of the system.

Structural and Behavioral Aspects

Structural Aspects focus on the static parts of the system. These include:

- Classes (e.g., Medicine, Customer, Billing)
- Attributes (e.g., medicineName, expiryDate, price)
- Relationships (e.g., association between Prescription and Medicine)

Behavioral Aspects deal with the dynamic behavior or interactions within the system. These include:

- Operations/methods (e.g., addMedicine(), generateBill())
- Interaction between objects (e.g., Pharmacist interacting with Customer to issue prescription)
- Sequence and activity diagrams (although not part of class diagram, they help describe behavior)

Class Diagram

A Class Diagram is a type of UML diagram that shows:

- Classes and their responsibilities (attributes and methods)
- Relationships among classes such as:
 - **Association:** A relationship where all objects have their own lifecycle and there is no owner.
 - **Aggregation:** A specialized form of association where objects have their own lifecycle but there is ownership.
 - **Composition:** A form of aggregation with strong ownership; if the parent object is destroyed, child objects are also destroyed.
 - **Inheritance (Generalization):** One class inherits attributes and methods from another.

Elements in Class Diagram

1. **Class:** A template that defines attributes and methods (e.g., Medicine)
 2. **Attributes:** Properties of a class (e.g., medicineName, batchNumber)
 3. **Operations/Methods:** Functions a class can perform (e.g., checkAvailability())
 4. **Relationships:** Lines connecting classes showing associations, dependencies, inheritance, etc.
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Case Study

Consider a pharmacy where a customer approaches the pharmacist with a prescription. The pharmacist checks the availability of the medicines, updates the inventory, generates a bill, and provides the medicines to the customer. This workflow can be effectively represented using UML class diagrams showing the interaction and structure.

Conclusion

This practical helped us understand how to model a Pharmacy Management System using UML class diagrams. We identified key classes, their attributes, methods, and relationships, gaining insights into the system's structure and behavior.

